Universidad Nacional Abierta y a Distancia
Vicerrectoría Académica y de Investigación
Course Natural Language Processing)
Code 203238430

Learning Guide - Phase 3 – Sentiment Analysis and Information Retrieval

1. Data of the Phase

Table 1. *Description.*

Aspect	Description
1. Type of activity	Independent
2. Evaluation moment	**Intermediate**
3. Managing unit	Escuela de Ciencias Básicas Tecnología e Ingeniería ECBTI
4. Score of the Choose an option	**120 points**
5. The activity starts on:	Sunday, September 27, 2026
6. The activity ends on:	Saturday, October 24, 2026
7. Independent work hours of the student	**32**

2. Detailed Description of the Learning Activity

Through this activity, the following learning outcome is expected to be achieved:

Learning Outcome 2: Discover relevant information from texts and categorize them using NLP techniques.

Structure data-driven solutions and make informed, efficient decisions that generate added value in the real-world contexts where the analyzed data are applied.

The activity **consists** of:

The practical component consists of two computational exercises and a technical report. In Exercise 1, you will work with a corpus of mental health–related statements to perform both unsupervised and supervised sentiment analysis, using a semi-supervised strategy to construct a labeled dataset. In Exercise 2, you will design, implement, and evaluate

an information retrieval (IR) model using a classic benchmark collection. Exercise 3 requires you to synthesize the methodology and results of both exercises in a structured article-style report appropriate for a master's-level course.

This guide explicitly defines two reference corpora, both available as CSV/text files on Kaggle:

- For Exercise 1 (Sentiment Analysis): the "Sentiment Analysis for Mental Health" dataset from Kaggle, hereafter referred to as the mental_health.csv corpus. This corpus contains mental health–related short statements tagged with mental health statuses or sentiment categories. Kaggle (***Annex 2 - Combined Data.csv***)
- For Exercises 2 and 3 (Information Retrieval and technical report): the CISI collection CISI: a dataset for Information Retrieval "***Annex 1 - dataset for Information Retrieval.zip***", a public benchmark dataset from the University of Glasgow's Information Retrieval Group. The collection includes 1,460 documents, 112 queries, and relevance assessments linking queries to relevant documents.

Exercise 1: Sentiment Analysis on the mental_health.csv Corpus

In this exercise, you will implement a sentiment analysis model in a Python Notebook using the mental_health.csv corpus (Sentiment Analysis for Mental Health dataset from Kaggle). Treat each textual instance in the dataset as a document expressing a mental health–related opinion or status.

The exercise is structured in three main stages:

1) Unsupervised sentiment analysis with lexicons

Select and justify the use of at least one unsupervised, lexicon-based sentiment analysis method. Possible options include, among others, Bing Liu, MPQA, Pattern, TextBlob, VADER, or lexicons specifically oriented towards mental health discourse. Make sure that the method you choose is appropriate for the language of the corpus (English).

- Apply the chosen lexicon-based method to the mental_health.csv corpus to assign a sentiment label or score to each instance.

- Briefly explain how the lexicon works (e.g., polarity scores, aggregation rules, handling of negation).
- Present a descriptive summary of the resulting sentiment distribution (e.g., proportion of positive, negative, and neutral instances).

2) Semi-supervised labeling strategy

Although the Kaggle dataset includes labels, for pedagogical purposes you will emulate a semi-supervised process. Proceed as follows:

- Randomly select a small subset of instances for manual inspection and create two disjoint subsets: one for training and one for testing (gold-standard labels). Use either the original labels or manually refined labels as your ground truth.
- Using the training subset, construct a model or apply a semi-supervised algorithm to propagate labels to the remaining unlabeled (or label-hidden) instances. You may choose from the following alternatives:

a) Train a supervised model (e.g., SVC, logistic regression) on the labeled subset and use it to label the remaining corpus.
b) Use a semi-supervised algorithm such as label propagation or label spreading.
c) Combine several models and build a more conservative corpus by keeping only instances where two or more methods agree on the sentiment label.

Use the manually labeled test subset to estimate how accurate your semi-supervised labeling process is, reporting appropriate evaluation metrics.

3) Supervised model and evaluation

Once you have a fully labeled corpus (original labels or labels generated by your semi-supervised process), train at least one supervised model for sentiment classification. You may choose any of the following algorithms, or others that you justify adequately:

- Support Vector Classifier (SVC).

- Logistic Regression.
- Naïve Bayes.
- k-Nearest Neighbors (k-NN).
- Decision Trees.
- Random Forests.

Evaluate the model(s) using the most common performance metrics for classification: accuracy, confusion matrix, precision, recall, and F1-score. When appropriate, complement these with macro- and micro-averaged versions of the metrics to account for class imbalance.

Finally, apply an interpretability method (for example, LIME, SHAP, or feature importance analysis) to at least three representative instances from the corpus. Discuss which words or patterns drive the model's decisions and how this relates to mental health–related content.

Exercise 2: Information Retrieval with the CISI Corpus

In this exercise, you will design and evaluate an information retrieval (IR) system using the CISI corpus. The collection contains 1,460 documents, 112 queries (CISI.QRY), Boolean query examples (CISI.BLN), and relevance assessments (CISI.REL) that indicate which documents are relevant to each query.

Your tasks are organized as follows:

1) Training and evaluating the retrieval model

- Load and preprocess the CISI.ALL file, segmenting it into individual documents.
- Load the 112 queries from CISI.QRY and the relevance judgments from CISI.REL.
- Implement at least one ranking model (e.g., vector space model with cosine similarity, BM25, or a neural ranking model).
- Evaluate how close your system's ranked lists are to the relevance judgments using standard IR metrics such as precision, recall, F1-score at different cutoffs (e.g., @5, @10), and, when possible, mean average precision (MAP) or nDCG.

2) Experiments with text preprocessing and representations

- Perform a careful preprocessing of the documents and queries (tokenization, case normalization, stopword removal, handling of punctuation and numbers, and optional stemming or lemmatization).
- Use at least two different document and query representation modes: a classical sparse representation such as TF–IDF, and a dense representation based on embeddings (e.g., Word2Vec, GloVe, doc2vec, or sentence-transformer embeddings).
- Enable queries to be issued in natural language text, mapping them into the same representation space as the documents.
- Compare the retrieval effectiveness of the different representations and ranking model combinations using precision, recall, F1-score, and any additional IR metric you consider relevant.

Exercise 3: Technical Report in Article Format

In Exercise 3, you will write a technical report that compares the techniques and results obtained in Exercises 1 and 2. The document should be written in clear, formal academic English and follow the structure of a short research article.

The report must include, at least, the following sections:

Introduction:

• Brief contextualization of sentiment analysis and information retrieval in the context of Natural Language Processing.
• Motivation for using a mental health corpus and the CISI collection.
• Objectives of the practical component.

Objective of the practice:

• Explicitly state the overall objective and specific objectives of the practical work.

Process – Exercise 1 (Sentiment Analysis on mental_health.csv):

• Description of the initial corpus: number and type of attributes, number of instances, source, and basic exploratory analysis.
• Data preprocessing and any feature or attribute selection performed.
• Construction of the labeled corpus (semi-supervised strategy and criteria used).
• Supervised model(s) trained and main hyperparameter choices.

• Results by model and a consolidated comparison table.
• Three examples of interpretability analysis and their implications.

Process – Exercise 2 (Information Retrieval with CISI):

• Corpus description: number and type of attributes, number of instances, and file structure (CISI.ALL, CISI.QRY, CISI.REL, CISI.BLN).
• Data preprocessing and attribute/feature selection.
• Ranking models used and their optimization process (hyperparameter search and validation strategy).
• Results obtained using each representation and retrieval model considered.
• Consolidated results comparing alternative configurations.

Conclusions and discussion – Exercise 1:

• Which models performed best and worst? Were there any particularly striking behaviors?
• Based on the reported metrics and interpretability analyses, to what extent could these models be used responsibly in real-life mental health–related applications?
• What additional aspects of the corpus, methodology, or results are worth highlighting?

Conclusions and discussion – Exercise 2:

• According to the evaluation results, is your IR model ready to be deployed in an organizational context? How close is it to the relevance judgments?
• What additional elements are worth highlighting regarding the process, results, or models used (e.g., robustness, computational cost, ease of interpretation)?

All written products must demonstrate rigorous academic writing, including a clear structure, logical coherence, appropriate cohesion between sections, and a formal style consistent with master's-level work.

References must follow current APA standards. You are strongly encouraged to use the Turnitin tool available on the virtual campus to review and improve the originality and academic integrity of your written work prior to submission.

3. Guidelines for the Development and Submission of Learning Evidence

The learning evidence are the actions, products, or observable processes that are carried out and/or submitted to demonstrate the capabilities, skills, and attitudes acquired by the student. These products also allow the tutor to verify and evaluate the student's performance.

For this practical component, the **individual deliverables** are:

- Participation in the discussion forum (Phase 3: Sentiment Analysis and Information Retrieval) Contribute to the forum corresponding to Phase 3 to engage academically with your tutor and classmates on the topics of sentiment analysis and information retrieval. Your contribution should not be a copy of the final report, but a meaningful and well-argued input that reflects your understanding of the exercises, methodological decisions, and preliminary findings.

- Computational notebook for Exercise 1 (Sentiment Analysis on mental_health.csv) Submit a Jupyter Notebook (or equivalent) containing the complete development of Exercise 1, including:

  ✓ Data loading and preprocessing steps.

  ✓ Unsupervised lexicon-based sentiment analysis.

  ✓ Semi-supervised labeling strategy.

  ✓ Training, evaluation, and interpretability analysis of the supervised sentiment model(s).
  ✓ The notebook must be clearly organized, with comment code, intermediate results, and brief textual interpretations.

❖ Computation notebook for Exercise 2 (Information Retrieval with the CISI corpus)
  ✓ Submit a Jupyter Notebook (or equivalent) containing the implementation and evaluation of the IR system, including:

  ✓ Preprocessing of documents and queries.

- ✓ Definition of document and query representations (e.g., TF–IDF and embeddings).

- ✓ Implementation of at least one ranking model.

- ✓ Evaluation using standard IR metrics and comparison of alternative configurations.
- ✓ As in Exercise 1, the notebook should be well structured, with commented code and concise explanations of the results.

❖ Technical report in article format (Exercise 3)
- ✓ Submit a written technical report that synthesizes and compares the methodology and results of Exercises 1 and 2, following the structure of a short research article (Introduction, Objectives, Process: Exercise 1 and 2, Conclusions and discussion, References).
- ✓ The document must be written in clear, formal academic English, use appropriate APA style references, and be delivered in PDF or Word format, according to the instructions on the virtual campus.

You are strongly encouraged to use the Turnitin tool available on the virtual campus to review and improve the originality and academic integrity of your written work prior to submission.

4. Academic Situations

Consider that in Agreement 029 of December 13, 2013, Article 99, the following actions are considered as offenses against academic order, among others: item e) "Plagiarism, that is, presenting as one's own the entirety or part of a work, paper, document, or invention created by another person. It also includes the use of false citations or references, or proposing citations where there is no match between the citation and the reference," and item f) "Reproducing or copying, for profit, educational materials or results of research products that have intellectual property rights reserved for the University."

The academic sanctions that the student will face are as follows:

a) In cases of proven academic fraud in the respective academic work or evaluation, the grade imposed will be zero points without prejudice to the corresponding disciplinary sanction.

b) In cases related to proven plagiarism in academic work of any nature, the grade imposed will be zero points, without prejudice to the corresponding disciplinary sanction.